NETWORKING.

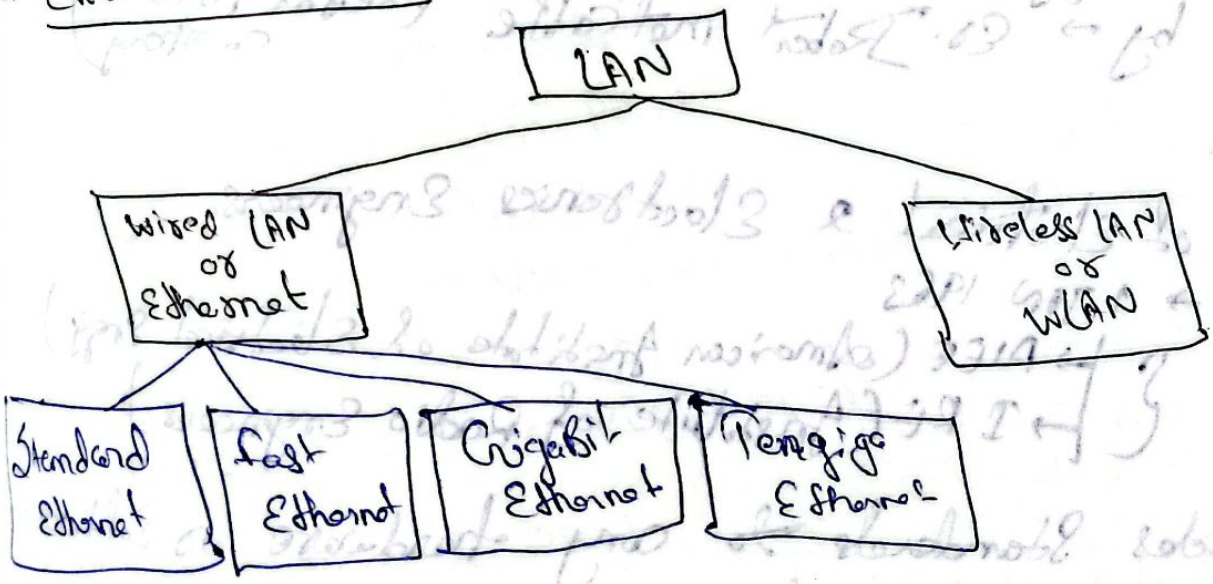
* Network :- It is a medium through which a connection is established b/w multiple computer. Wired or wireless.

* Types:
(1) LAN, (2) CAN, (3) MAN, (4) WAN, (5) PAN, (6) SAN ... etc.

1) LAN (local Area Network):-

→ Private and Secure
→ Range upto 2km.
→ Small geographical area like School, office...
→ All resources are connected to a central device. (hub or switch)
→ A router is used to connect the LAN to the Internet.
→ To create a wireless LAN, a WAP (wireless Access Point) is used.

* Evolution of LAN :-



Types of LAN :-

(1) ARCNET :- Attached Resources computer Network.
↳ Oldest and Easy installation
↳ Started 1977 by data point corporation - New York
↳ 1980 - available for microcomputer. Speed - 2.5 Mbps.
↳ Token passing mechanism

(2) CAN (Campus Area Network) :-

→ Range 1 to 1km.
→ LAN < CAN < MAN.
→ connection of Inter-connected LAN.

(3) MAN (Metropolitan Area Network) :-

→ Big Geographical Area use. like a City.
→ work on IEEE 802.6
→ wired MAN uses optical fibre. (work on 802.8)
→ use Data link layer protocol.

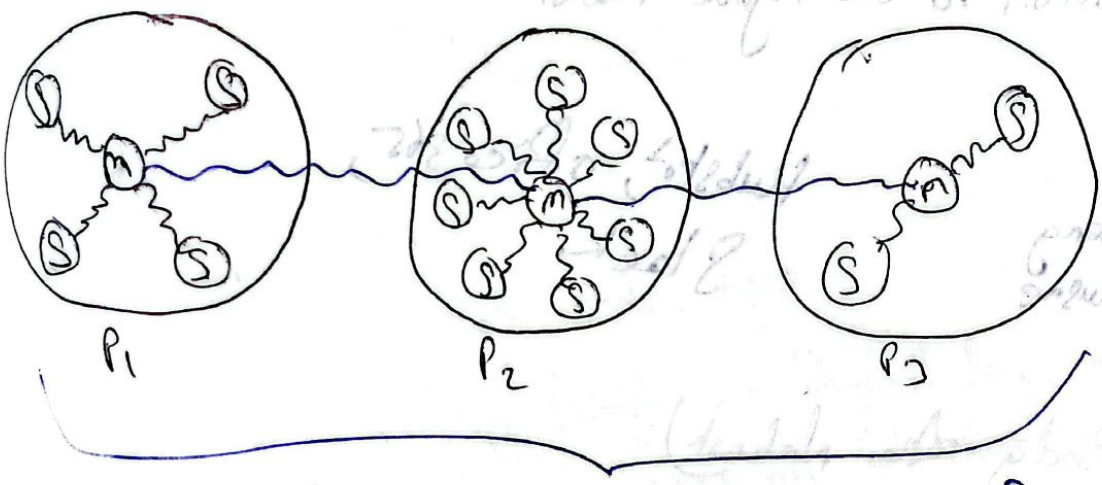
(4) WAN (wide Area Network) :-

→ largest network of the world. (known as Internet)
→ least Secure & low Speed.
→ Network → to Network.
→ use VLPS (variable length package Switching) for Communication
(work on Network layer)
→ use fibre optics Cable.
→ use Satellite.
  - Repeaters — Transponders.
  - uplink — Earth → Satellite.
  - downlink — Satellite → Earth.

(5) PAN (Personal Area Network) :-

→ known as WPAN
→ work on IEEE 802.15
→ Bluetooth Connectivity.
  - IEEE 802.15.1
  - Range - 10 m.
  - Started by Jaap Haartsen
  - Adhoc Network
  - 7 devices connect.
    - master Device (1)
    - Slave Device (6)

→ only master can send data.
→ master & slave can connect with each other.
→ standard transmission rate — 720 kbps
→ works on 2.4 GHz frequency
→ low cost & short range.
→ Basic Architecture of Bluetooth is known as PICONET
→ Group of PICONET known as SCATTERNET.
→ Master to master connection needed to connect multiple PICONET.
→ Can be use slave, which act as Bridges.



SCATTERNET (3M, 13S)

⑥ SAN ⇒ (Storage Area Network).
→ known as Cloud Services.
Started : 1999. (by Salesforce (saas)
Amazon = AWS.
Apple = Icloud
Microsoft = Azure
Google = GCP.

* Cloud Service model:

(1) **IaaS** :-
→ Infrastructure as a Service
→ Max. flexibility
→ High availability
→ Cost and time saving.

Azure, AWS, GCP

(2) **PaaS** :-
→ Platform as a Service
→ Cost effective development.
→ High availability.
→ Ease migration to the hybrid model

Docker, Kubernetes.

(3) **SaaS** :-
→ Accessibility
→ Cost efficiency
→ All-Inclusive

Hubspot, office 365, Slack.

(7) **BAN** (Body Area Network) :-
→ It Bug/chip kit to person body to trace.

(8) **HAN** (Home area network)
→ wifi modem to connect home/office Devices.

(9) **VPN** (Virtual Private Network).
→ To extend ~~public~~ private and secure network to public network
→ Work from home.
→ Encrypted Connection, Blocked website, Data secured.
→ Change IP.

* P2P Network (Peer to Peer).

→ Decentralize network (because No Server)
→ Transfer bigger data easily
→ Each computer work as client & Server.
→ Started 1979.

Advantages :-

→ No need of any OS.
→ If any computer failed, No issue in Network.

Dis-advantage :-

→ No Central Server.
→ No Data Back-up.
→ No Security, Slow Network.

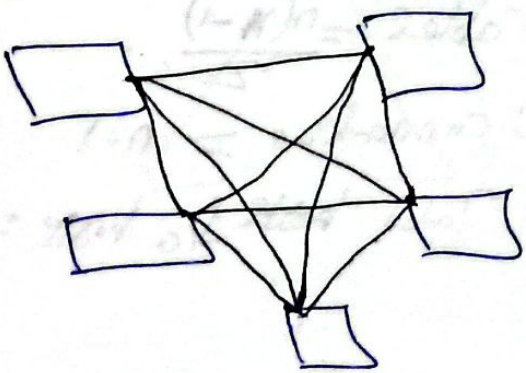
* Network Topology :-

→ Based on connection, two types.
(a) Physical Topology (works on physical layer of OSI model)
(b) Logical Topology (works on network layer of OSI model)

Types:

(1) Mesh :-

→ Every node connected to each other.
→ Data flow - bi directional
→ use in WAN.
→ fully connected Topology.
→ Most reliable & Secure.
→ Decentralised network



→ Connected with channel / link / cable.
→ Redundancy & fault tolerance.
→ multiple path b/w two nodes.

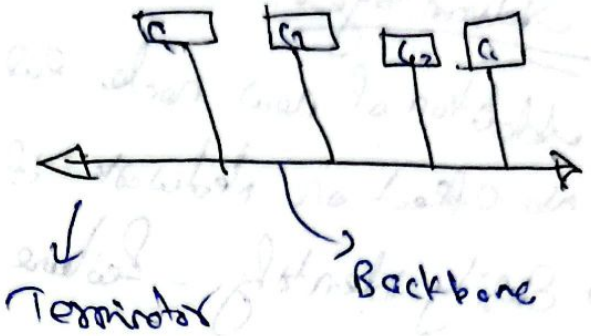
* Types of mesh topology :-
(i) fully mesh → nodes fully connected, (n-1) connecting
(ii) Partial mesh → nodes not fully connected, use optional path.

* Advantages :-
→ one device fail, no network failure.
→ no traffic problem.
→ high privacy and security.
→ Data transmission is more consistent because any fail node/cable doesn't interrupt the transmission.
→ Addition of new device will not disturbt the data transmission.

* Disadvantages :-
→ Bulk cabling required.
→ Installation and configuration is high.
→ Most Extensive.
→ Each node requires extra utility cost.
→ More maintenance required
→ Higher Power req.

(a) cables $= \frac{n(n-1)}{2}$
(b) connections $= n-1$
(c) Total I/O ports $= 2 \times$ total Cables.

(2) Bus Topology :-



- Connected with a single cable
- known Bus / Backbone / Segment.
- Line / linear / Horizontal.
- Data flow unidirectional.
- max = 30, Node can connect.
- Terminator / resistor at the is connected both the end to block Data loss.
- Ethernet (wired LAN use.

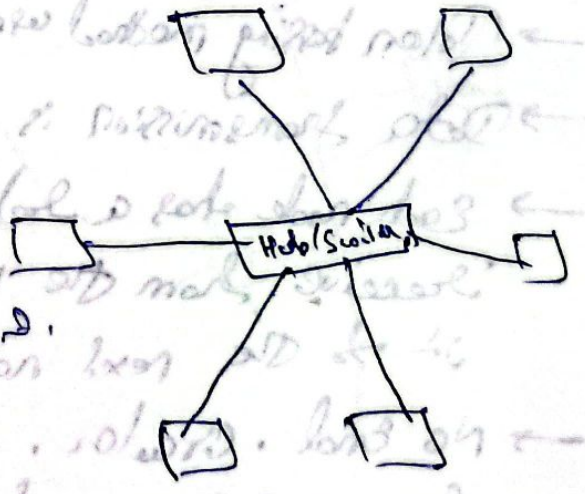
* Advantage :-

- Simple & inexpensive
- Scalable - Addition of new device easy.
- Total cable required = n+1

* Disadvantage :-

- If backbone fail, all network fail.
- Limited length of Network.
- Data collision.

(3) Star Topology :-



- Data Broadcast.
- Hub/switch - control device which share connection point to point.
- Hub, OSI model (Physical layer)
- Data flow - unidirectional.
- Cables = No of nodes.

* Advantages:

→ Addition of new node easy
→ No effect on network if - one node fail.
→ Easily identify - failure node.

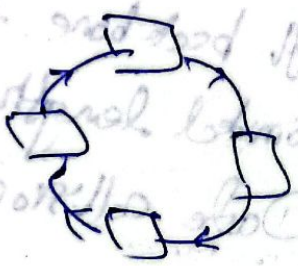
* Disadvantages:

→ Central Server fail - whole network fail.

* Types:

(a) Broadcast Star Network :-
→ Hub used for centralized devices
(b) Switched Star Network :-
→ Switch used, modern Switch ~~used~~ both unicast & multicast

(4) Ring Topology :-
→ Each node is connected to the next and prev node - forming a ring
→ Data passes through every node.
→ If any single node fail - whole network fail.
→ Data flow uni-directional.
→ Token passing method used for data transfer.



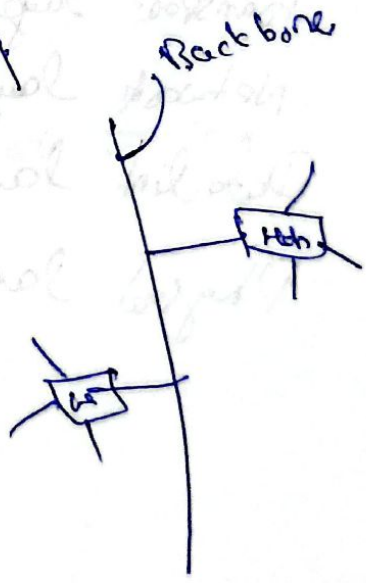
→ Data transmission is bitwise.
→ Each node has a repeater that amplified the data receive from the previous node and then send it to the next node.
→ No End. Circular.
→ RJ-45 or co-axial cable use.
→ No need of any centralize device

* Advantages:
- Faster speed, Reliable, Expansion easy
- No collision, because of unidirectional.

* Disadvantage.
- Slower than Star topology.
- more node, less speed.
- one node fail, whole Network fail.

* FDDI :- (Fibre Distributed Data Interface)
→ Bidirectional.
→ Contain two rings - Primary & Secondary.
→ Primary - Data.
→ Secondary - Backup data.
→ max length = 200 km.
→ max node = 1000 nodes.

5) Tree Topology :-
→ combination of Bus and Star topology.
→ Hierarchy topology.
→ Use in WAN.
→ Extended version of Star topology.
→ Backbone fail, not all network fail.
→ Secure topology.
→ Best performance than Ring.



* Advantages:-

→ Error detection & correction easy.
→ Secure and Reliable.

* Disadvantages:-

→ Setup cost high
→ Configuration tough
→ More cable use.

* Networking devices:-

→ The devices which is used to connect client & servers.

Application layer → Gateway, Firewall
Presentation layer → Gateway,
Session layer → Gateway,
Transport layer → Gateway, Packet filter firewall.
Network layer → Gateway, Router, Brouter (Bridge + Router), 3-layer switch
Data link layer → Gateway, Bridges, NIC, 2-layer Switch
Physical layer → Gateway, Modem, Hub, Repeaters, Cables.

* Networking Devices :-

1) MODEM :-

- modulator - Demodulator
- Analog to digital ⇒ Demodulator
- Digital to Analog ⇒ modulator.
- Use in Star Topology
- Physical layer
- Telephone lines & Computer use
- 2 modems connects each other called Handshaking.
- Speed = BPS (Bits per sec).
- Also called as Hybrid device,
- Work as both input and output device

2) Repeaters :-

- Amplify & Regenerate weak signals.
- Physical layer -
- Use ~~for~~ to extend length of LAN, or to connect two separate LAN.
- Non-Intelligent Device.
- ~~Repeat~~ Pure / Dumb Hardware.
- Only forwarding, No data filter
- Hubs called as multiport repeaters.

(3) Hub :-

→ works on Physical layer
→ Multiport Repeater.
→ Non-Intelligent device.
→ from one node → to all connected nodes broadcast

* Types :-

(a) Passive :-

→ act like a connector only.
→ Can't work without electricity.
→ work below Physical layer.
→ Doesn't Regenerate Signals

(b) Active Hub :-

→ works as a Repeater.
→ Mostly used basic hub in any element.

(c) Intelligent Hub :-

→ Type of Active hub, but Regenerate & monitor Signal
→ use SNMP.
→ Also known as Switch.

(4) Switch :-

→ Intelligent device (knowledge of MAC)
→ Data Transfer → to Specific Receiver
→ work on Data-link layer.
→ 5-52 ports.
→ To forward the data → to Receiver, in Switch a table, CAM (content Accessible memory).

Topic:

(4) 2-layer Switch:-
→ work on Data link layer of OSI model.
→ Multiport Bridge (also called)
→ work on MAC address.

(4) 3-layer Switch:-
→ work on Network layer.
→ called as Router.
→ work on IP address.

(5) NIC (Network Interface Card):-
→ Hybrid device.
→ known as, Physical address, MAC address, Hybrid address, Burned in Address, Chesis No., Data link address.
→ 6 bytes / 48 bits / 12 Hexadecimal digit.

__:__:__:__:__:__

→ 5 Seperators (:).

(6) Routers:-
→ connect LAN to WAN.
→ work on Physical, Data link & Network layer.
→ It created a routing table for all possible receivers.
→ It works as traffic director for data packets
→ Use IP address.
→ Connect two different Topology.
→ Use Routing Information protocol (RIP), OSPF (open shortest path first), BGP (Border gateway protocol).
→ Use firewall for unauthorised Access.

(7) Bridges :-

→ work on data link layer of OSI model.
→ used for extend LAN.
→ Can do data filteration
→ Use MAC address to connect LAN.

(8) Gate way :-

→ work throughout the OSI model & TCP/IP model.
→ Also known as firewall.
→ Connect all LAN of different configuration.
→ work as Entry and Exit point.
→ Protocol Converter.
→ Router works with Network but Gateway facilities to work with different model.

(9) Brouter :- (Bridge + Router)

* OSI model :-

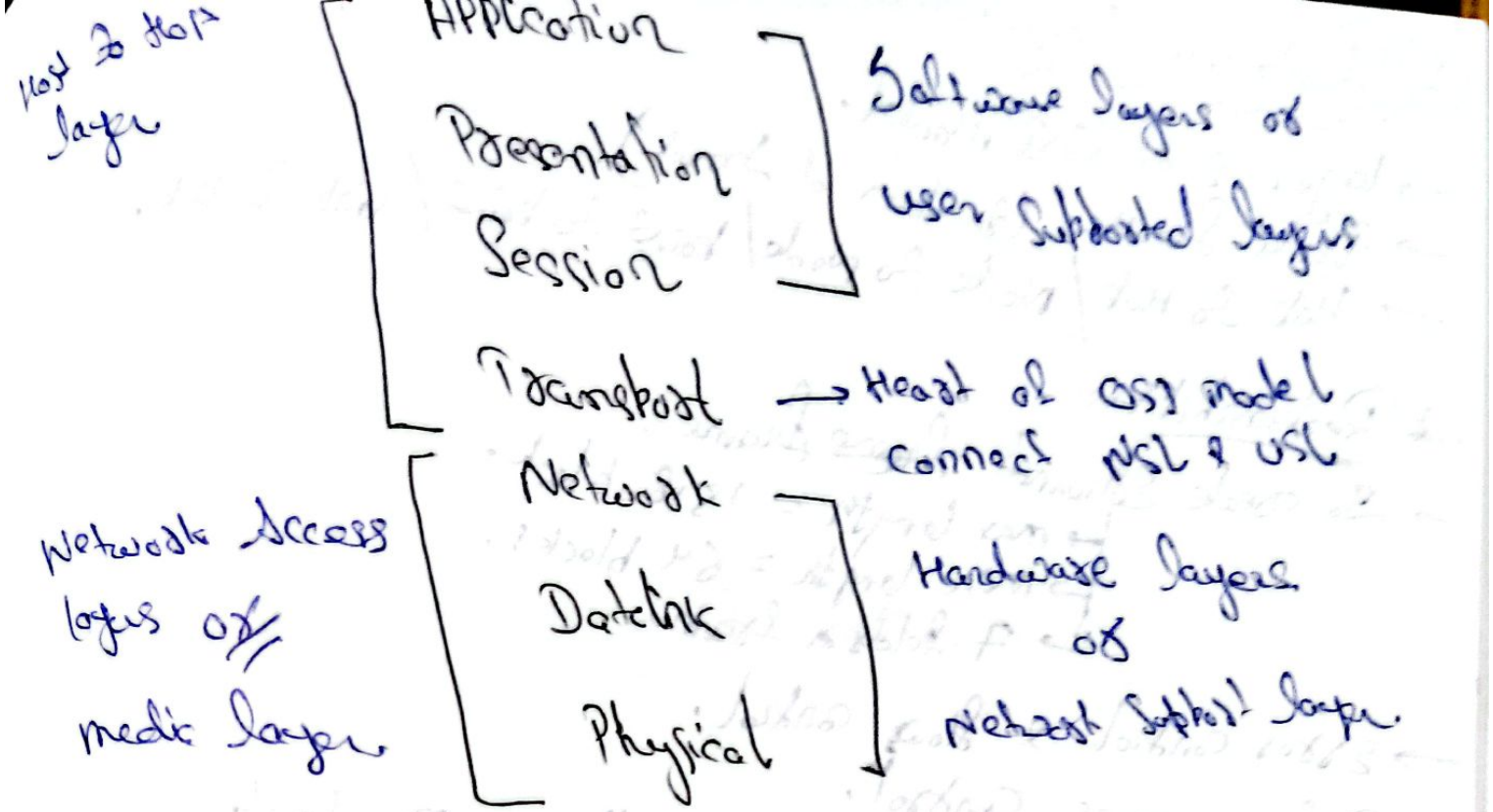
→ Open System Interconnection
→ Also known as.
Referential model, theoritical, logical, Conceptual model.
→ 1970 by Charles Bachman.
→ Published by ISO (1984)

Please do not take Sabji.

Pooja Acharya.

Data flow.

7 Application layer → DATA / MESSAGE
6 Presentation layer → DATA
5 Session layer → DATA
4 Transport layer → Segment / Datagram
3 Network layer → Packets
2 Data link layer → frames
1 Physical layer → BITS.



1) Physical layer:-

→ 1st layer, Data bits travel.

* Responsibility :-

→ Defines physical topology.
→ Defines data transmission modes. (simplex, duplex, half duplex)
→ Defines transmission media.
→ Synchronise bits to balance source and destination.

* Limitations :-

→ No error handling, No data interpretation.
→ Susceptible to physical damage.

* Protocols:

1) ISDN (Integrated services digital Network)
2) SONET (Synchronous Digital Network)
3) Bluetooth, Hub, gateway, Repeaters, modem, wifi, Infrared.
4) Ethernet, USB.

(2) Data link layer:-

→ layer 2 of OSI model.
→ Data frame (In form of bits)
→ Hop to Hop / Node to Node / point to point / link to link.

* Responsibilities:-
→ To create Ethernet ~~frame~~ frame.
  → max length = 1518 bits.
  → min length = 64 bits.
  → 7 fields in frame
→ Error control & flow control.
→ Define access control.
→ Use MAC address. (Unicast, Multicast, Broadcast

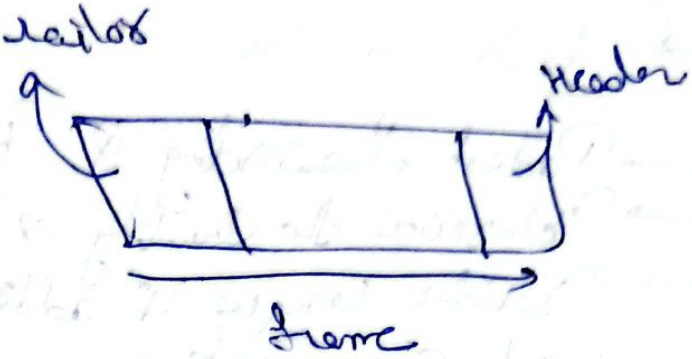
Q. which address is unicast?

(A) B6:A6:C2:B5:A2:B5 unicast.
(B) A5:B6:A2:B5:C2:B5
(C) E3:B5:A2:B6:A5:B5
(D) C5:A2:B6:B5:A2:B5
} multicast

* If the second digit of the first bit of MAC address is even number, then the address is UNICAST.
* If odd then MULTICAST.
* If 0 UNICAST.
* If all physical address f broadcast
  like ff:ff:ff:ff:ff:ff

* Error control in layer 2:-

- Add extra bits in frame to control error.
- Slot layer duplication of frame.



* Limitation of layer-2:-

- Limited Scale
- Increased overhead
- Error handling dependency. (For complex errors)
- No routing Capabilities

* Protocols:-

ARP (Address Resolution protocol).
DTP (Dynamic Trunking protocol).
LACP (Link Aggregation Control protocol)
PPP (Point to Point protocol)
HDLC (High label data link control protocol)
SDLC (Synchronous Data link protocol)
SLIP (Serial line Interface protocol)
LCP (Link Control protocol)
NCP (Network Control protocol)
Ethernet, FDDI, 2-layer Switch, Token Ring, Bridge, NIC, WAP.

3) Network layer:-

- Travel, In form of Data Packet.
- Source to destination / Host to Host deliver.
- Choose best path to deliver data.

* Responsibilities :

→ Packet forwarding & Routing.
→ Determine the Quality & priority of Packet being transmitted
→ Provides unique IP address to devices for identification and communication.
→ Send packets to the destination using routers.
→ Divides large data packet into smaller fragments for transmission and reassembles them at the receiver's end.
→ Divides larger network into smaller subnetwork (subnetting)

* Limitations :-

→ No flow control.
→ Routers may loose packets.

* Protocols :-

ICMP (Internet control message protocol)
EGP (Exterior Gateway Protocol)
IGMP (Internet group management protocol)
IGRP (Internet Gateway routing protocol)
RIP (Routing Information Protocol)
VRGP (virtual router Redundancy protocol)
IP-V4 & IP-V6.
RARP (Reverse address Resolution protocol)
OSPF (open Shortest path first)
BGP (Border Gateway)
IPSec (Internet protocol Security)
MPLS (multi protocol label Switching).

Transport layer:

- layer 4 of OSI model.
- Data level form of Segments / datagram.
- Heart of OSI model.
- link NSL and USL.
- Data actual Decomposition at Receiver End.
- Deliver data End to End / process to process / machine to machine.

Responsibility:

- Connection Control b/w Source & destination.
- Segmentation and Reassembling.
- Error control & flow control.
- multiplexing & Demultiplexing.
- Add Receiver port no. in data header (Service point addressing)
- TCP & UDP.
- End to End Connection control b/w ~~host~~ hosts.
- Congestion control.
- Data Integrity.

TCP	UDP
Transmission Control protocol	User Datagram protocol
Connection oriented	Connection less
Reliable & Secure.	Unreliable & unsecure.
Slow	fast
Doesn't support broadcasting	Supports Broadcasting.
Byte oriented	message oriented
Heavy weight	light weight

* Protocols:

TCP, UDP, FCP (Fibre channel protocol).
DCCP (Datagram Congestion Control protocol)
SCTP (Stream Control Transmission protocol)
ATP (Apple talk transaction protocol)
BGP (Border Gateway protocol)
SPX (Sequenced Packet Exchange)

(5) Session layer:-

→ layer 5.
→ Responsible for connection / session.

* Responsibilities of

→ Creation of connection, management & Termination.
→ Token management.
→ Data transmission mode
→ Synchronization is implemented by inserting checkpoints for recovery of the time of disruption
→ It ensures orderly communication
→ Dialogue control.

* Protocols:

NCP (Network Control protocol)
RP (Remote Control)
SIP (Session Initiation protocol).
PPTP (Point to point tunneling protocol)
ZIP (Zone Information protocol)
NFSP (Network file System protocol)
PAP (Password Authentication protocol)

RTCP (Real time control

6) Presentation layer:-

→ layer 6.
→ change data to standard format.
→ Concerned of Data Syntax & Semantics.

* Responsibilities:

→ Data Encryption & Decryption.
→ Data Compression & Decompression.
→ Data translation at destination

* Protocols:

SSL (Secure Socket layer)
JPEG (Joint photographic Experts Group)
MPEG (moving picture Experts Group)
TLS (Transport layer Security)
LPP (Lightweight presentation)

7) Application layer:-

→ layer 7.
→ Nearest to the user
→ Desktop layer.

* Responsibilities:

Email Related Services.
Remote login services.
file transfer & manage (FTAM)

* Protocols:-

AMQP (Advanced message Queuing protocol)
Bootp (Bootstrap)
DNS (Domain name system)
DHCP (Dynamic Host configuration protocol)
FTP
HTTP, HTTPS
IMAP (Internet message Access Protocol)
MIME (Multipurpose Internet mail Extention)
NTP (Network time protocol)
POP-3 (Post Office protocol v-3)
Telnet (Telecommunication Network)

* IP Address:-

→ Internet protocol.
→ logical Address.
→ work on Network layer.
→ Network ID + Host ID.

* management of IP Address:-

→ IP address managed by IANA (Internal assigned Number authority)
→ IANA IP address & port Numbers provide.
→ Administration RIR (Regional Internet registry)
→ 5 RIR:-

(a) APNIC (Asia Pacific Network Information Centre)
→ Start - 1993.
→ Cover - South, East & South-East Asia.
→ In India also.

(b) AFRNIC :- (African Network information Centre)
→ Start 2004.
→ Cover all Africa Continent.

(c) ARIN (American registry for Internet Numbers)
→ Start 1997
→ Cover USA, Canada, Antertica & Carribean.

(d) LACNIC (Latin America and Carribean Network Information Centre)
→ Start 1993
→ Cover Latin America & Carribean.

(e) RIPE-NCC (Resaux IP European Network coordination Centre)
→ Start 1992
→ Cover Europe, Central Asia, Westerlern Asia.

* Types of IP Address :-

(1) Public IP :-
→ Provided by ISP to Router.
→ to Direct Internet Access.

(2) Private IP :-
→ Networking devices. like router provided.

(3) Static IP:-

→ Static, Don't change, fixed IP.
→ Assigned by DHCP server.

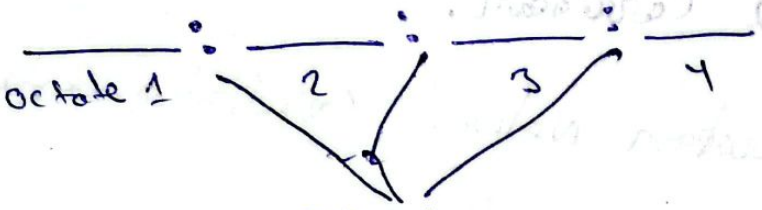
(4) Dynamic IP:-

→ Assign by ISP.
→ Dynamic IP DHCP & PPPoE (Point to Point over Ethernet)

* Version of IP:-

(A) IP-V4 ⇒

→ Internet protocol version-4.
→ Created in 1981
→



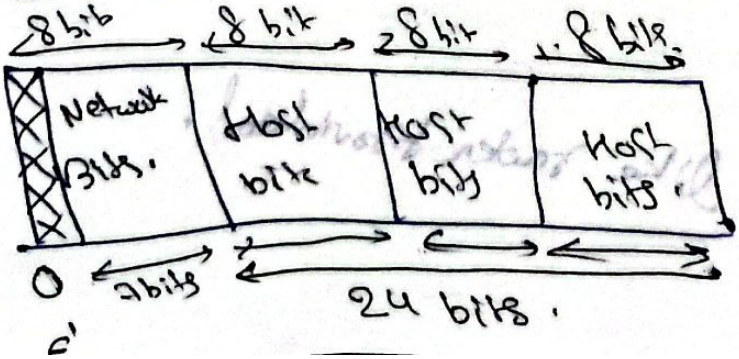
→ Size :- 8x4 = 32 bits.

max address → 2^{32}.

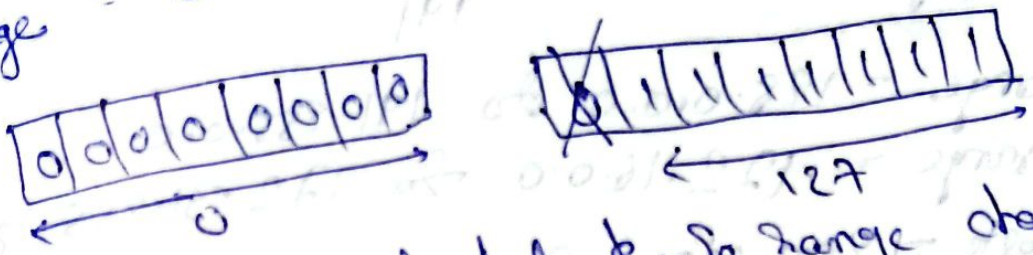
→ Unicast, multicast & Broadcast.

(1) CLASS-A :-

→ maximum number of hosts.

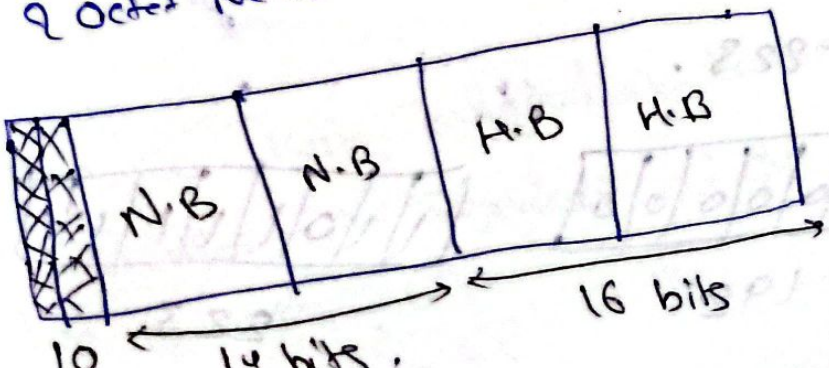


→ The first bit of first octate is 0, and remaining 7 bits combate the network ID.
→ By using the first octate as the network ID is class A, a total of 126 networks can be created.
→ Range = 0 to 127.



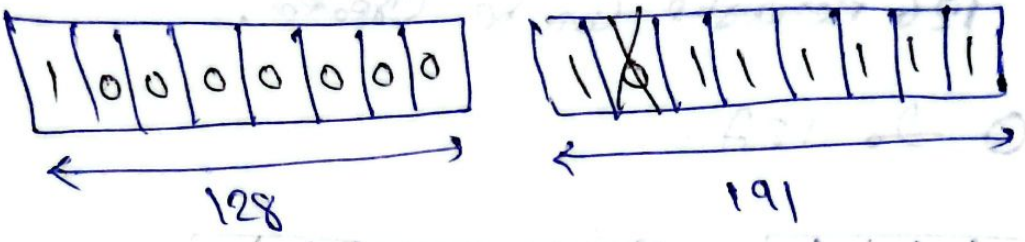
→ 127 reserved for local loop. So range change to 0-126.
→ Public IP range = 1.0.0.0 to 177.0.0.0
→ Private IP range = 10.0.0.0 to 10.255.255.255.
→ Total networks ⇒ $2^n - 2 = 128 - 2 = 126$ (n = network bits)
→ Total hosts ⇒ $2^h - 2 = 2^{24} - 2 = 16777214$. (h = host bits)
→ To get the Subnet mask, make all network bits 1 and host bits 0.
→ class A Subnet mask = 255.0.0.0.
→ use for unicasting.

(2) Class-B :-
→ Medium to large scale network.
→ Donate 2 octet Network bits, last 2 octet Host bits.



→ First 2 bits of N.B ~~are fixed~~ 1 and 0.

→ Range - 128 to 191



→ Public IP range = 128.0.0.0 to 191.255.0.0

→ Private IP range = 172.16.0.0 to 172.31.255.255.

→ Subnet mask = 255.255.0.0

→ Unicast.

→ Total network = $2^n - 2 = 2^{14} - 2 = 16384$

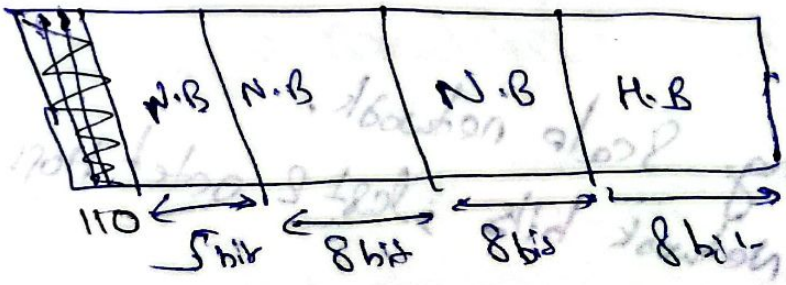
→ total host = $2^h - 2 = 2^{16} - 2 = 65534$.

(3) CLASS C:-

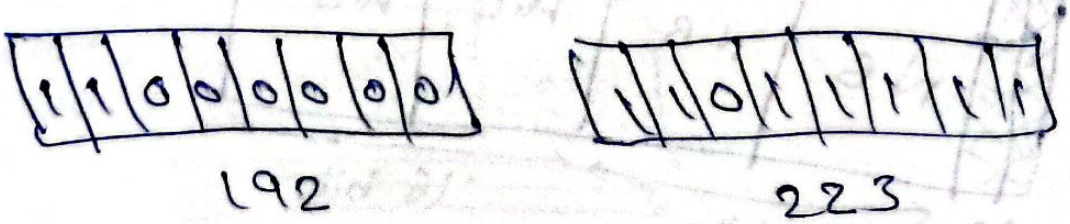
→ LAN or Small Geographical use.

→ 3 octet Network bit, Left octete Host Bits

→ first 3 bits of network bits Denote. Fix → 1, 1, 0.



→ Range = 192 - 223.



- Public IP range = 192.0.0.0 – 223.255.255.0
- Private IP range = 192.168.0.0 – 192.168.255.255.
- Subnet mask = 255.255.255.0
- unicast use,
- Bit network = $2^n = 2^{21}$.
- Bit host = $2^h - 2 = 2^8 - 2 = 254$ hosts.

(4) CLASS - D :-

- ~~All bits~~ only network bits, no host bits
- 4 bit fixed 1,1,1,0.
- Range = 224 to 239.
- Public IP range = 224.0.0.0 – 239.255.255.255

(5) CLASS E :-

- Not in use
- Reserved for Research purpose.
- 5 bits reserve 1,1,1,1,0.
- Range - 240 - 255.

* IP Address Configuration :-

(A) Automatically (B) Manually.

Protocols

(1) DHCP (Dynamic Host Configuration protocol)

- This protocol uses UDP services of the Transport layer.
- Server host number - 67
- Client host number - 68.

(2) WINS (Windows Internet Name Services).

→ When DHCP Server is unavailable, the OS uses an important feature called APIPA (Automatic Private IP Address)

→ APIPA range :- 169.254.0.1 to 169.254.255.254

* IPV6 :-

→ Created 1995

→ IETF (Internet Engineering Task Force)

→ Uses 8 octet, 8 x 16 = 128 bits. (Hexadecimal)

→ 7 seperators.

* TCP/IP model :-

→ Transmission Control Protocol / Internet Protocol model.

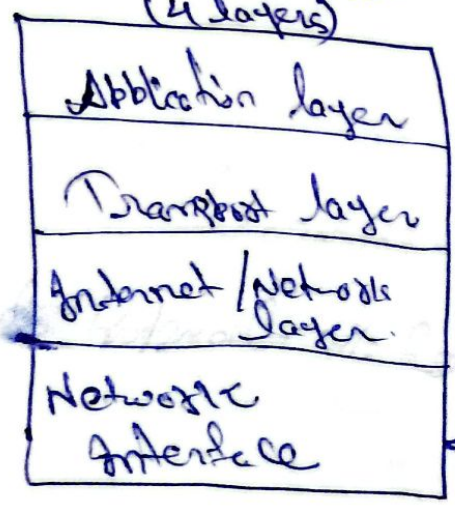
→ Work on Transport layer.

→ known as DoD model (Department of Defense)

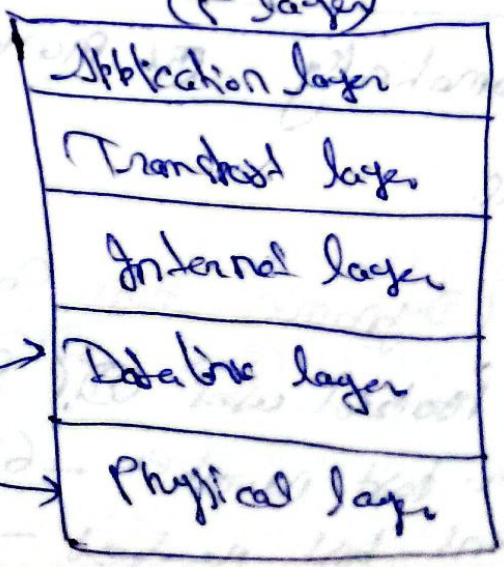
→ Developed by ARPA.

→ First TCP/IP the OSI model.

Original TCP/IP model (4 layers)



Modified TCP/IP model (5 layers)



* Data Communication:-

→ Exchange of Data b/w two devices by using communication medium.

→ Data communication - depends on 4 characteristics.

(a) Accuracy (b) Timeline (c) Delivery (d) Jitter (To deliver packet to receiver end)

* Components of Data Communication :-

(a) Sender (b) Receiver (c) Data & message (d) Communication medium

(e) Protocols (3 keys Elements)

(i) Syntax (ii) Semantics (iii) Timing (Speed of data to send)

* Data Transmission model:- (Responsibility of Pysical layer)

(A) Simplex :-

→ Unidirectional

→ Sender → to receiver

→ Sender can only send data but can't receive.

→ Receiver can only receive data but can't send.

Eg: Monitor, Keyboard, Mouse etc.

(B) Half Duplex :-

→ Two way communication possible but unidirectional at one time.

Eg: walkie-talkie.

(C) Full Duplex :-

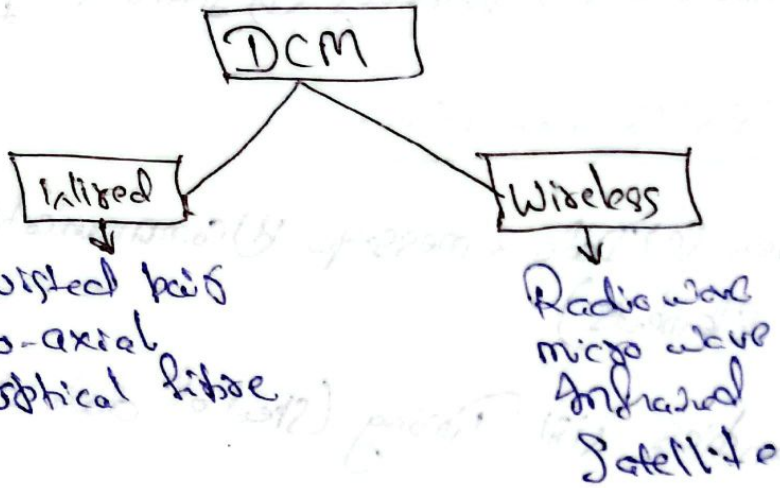
→ Bidirectional communication at a time.

→ Eg: Mobile, Telephone, Switch, TCP etc.

* Data Communication Media :-

→ The way or medium by which data communication take place.

Types :-



* Wired :-

(1) Twisted pair :-

→ When two seperated insulated wires are twisted around each other in a twisted pattern.

→ These two wires form a circuit through which data is transmitted.

→ When data flow in a wire, a magnetic field is created around it - the magnetic fields of the two wires cancel each other, which provides protection from crosstalk

→ Crosstalk :- The noise generated in a nearby wire.

(a) UTP (unshield twisted pair)

→ Speed 10-1000 mbps.

→ Use AWG (American wire gauge)

→ Insulated with PVC (Polythene) or FEP (Fluorinated Ethylene Propylene)

wires are color coded.
each color shows which application/use of wire is for
UTP color pairs :- Blue-white, Orange-white, brown-white, Green-white.

Advantages :-
cheap and easy installation
fastest copper medium.

Disadvantages :-
not used for long network.
crosstalk high.

(b) STP (Shield twisted pair)
Extra metallic-foil Shield.
Protected from interference coming from inside or outside.
Both ends must be grounded, otherwise signal leaks.
use 8-Pin RJ-45 connector.

(2) Co-axial Cable :-
first type of cable used in network. (Dish tv)
one copper wire (main conductor), covered by a solid electrical insulator
Insulator covered by foil and metallic mesh.
metallic mesh (outer conductor)
BNC (Bayonet Neil-Concelman) connector used.

Types.

(a) Thin net :-
follow IEEE 10Base2 standard.
max length - 185 m.
speed 10 mbps

→ can transmit only a single frequency, so it is called baseband.
→ BNC-T connector, which attaches to NIC.

(b) Thick Net :-
→ Based on IEEE 10base2 standard.
→ max-length 500 m.

(3) ofc / foc (optical fibre optics).
→ Contain very thin glass/fibre filaments through which data travels as a light beam.
→ To generate this light beam, LED or ILD.

Components:

(a) Core :-
→ made by glass fibre
→ Pass light beam.
→ main conductor
→ Data travels here.

(b) Cladding :-
→ Protect core
→ flexible plastic

(c) Coating :-
→ outer layer
→ Plastic

→ A layer of polymer / polymide is applied over the cladding
→ It converts coded electronic pulses into light pulses and transmits them.
→ At receiver end light pulses are converted back into binary data by a Photoreceptor.
→ Data travel using total internal reflection.

* modes of ofc :-

(a) Single mode : (Costly, High bandwidth, Small core)
(b) multi mode :- (cheap, Low Bandwidth, Bigger core)

* Wireless / unguided media :-

(1) Radio waves :-

→ Frequency 10 KHz - 1 GHz.
→ Bypass obstacles.
→ Types :

Short range = (AM)
VHF (very high frequency) = FM radio
UHF (Ultra high ") = TV.
SHF (Super high ") = 3 - 30 GHz.

(2) Microwave :-

→ Use special type of Antenna. which send signal to other th.
→ work on line of sight.

(3) Infrared :-

→ Second wireless communication.
→ Used in Remote, wireless keyboard.
→ Pass obstacle, small area.

(4) Satellite :-

→ Special case of microwave relay.
→ Transponder use.
→ Uplink ↑ Downlink ↓

* Backup & Restoring Data:-

(A) RAID Configuration:-

→ Redundant Array of Independent Disks.
→ In network, data is stored across multiple disks.
→ Provide fault tolerance.
→ Performance & Reliability.

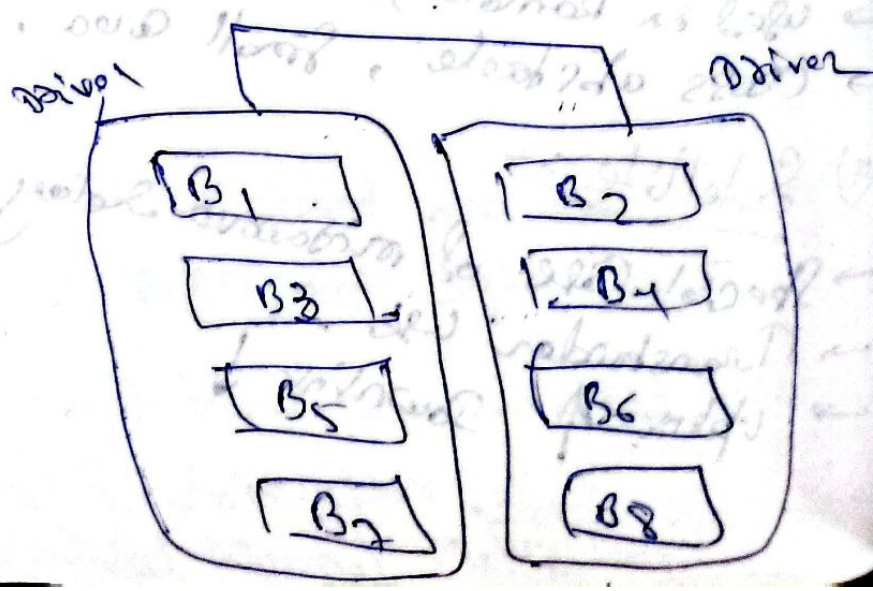
* Data Striping:-

→ Storing bits of data on different hard disks is called data striping.
→ It used for RAID level 0 onwards.
→ Can be 3 types:- Bit level, Byte level & Block-level.

(1) RAID level 0:-

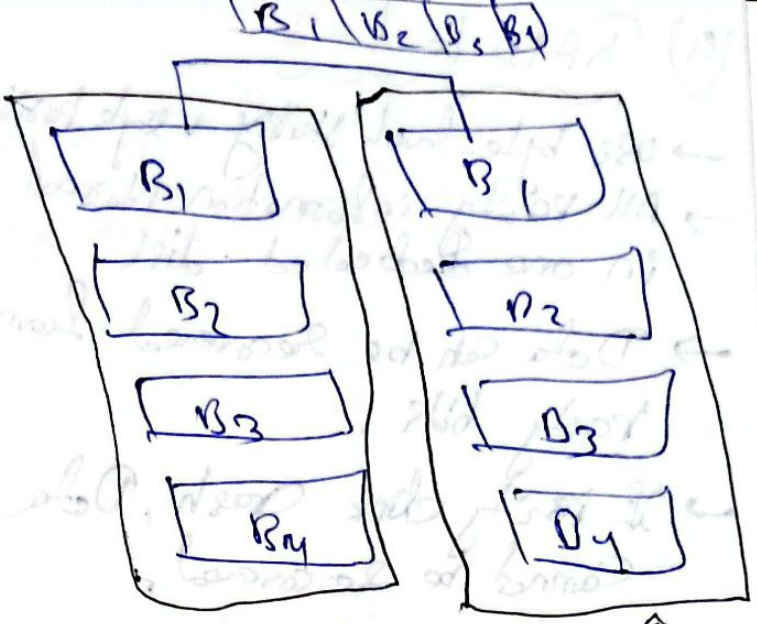
→ Uses block level striping.
→ Data blocks are stored across multiple disk.
→ If any disk/data lost, can't be recovered.
→ Gives highest/maximum performance (100%) and is the fastest RAID level.
→ No parity bit is used.
→ No fault tolerance.





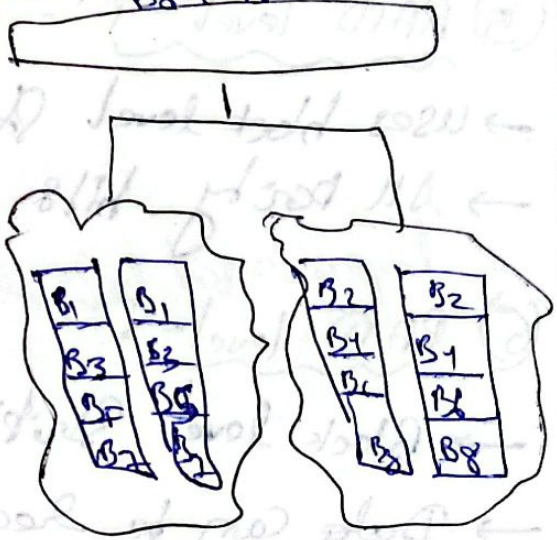
(2) RAID level 1 :-

- shadowing / cloning / mirroring
- Block level Stripping
- min. 2 disks are used.
- Performance 50%.
- No parity used
- Storage mirror-set.



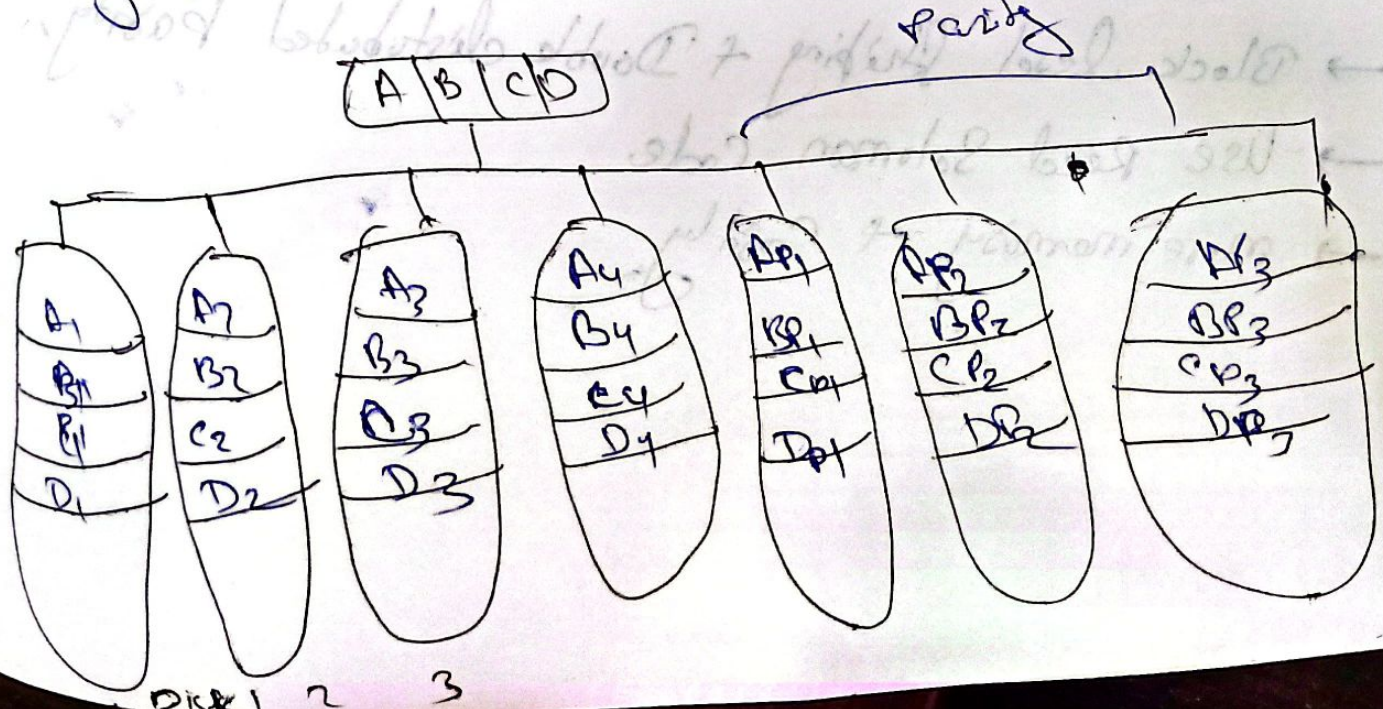
(4) RAID level -10 :-

- RAID level 1 + RAID level-0
- Nesting
- Block level Stripping & mirroring.
- Provides performance & reliability
- No parity used.



(3) RAID level-2 :-

- Bit level Stripping + ECC (Error correcting code)
- Parity bits are used to detect & correct errors.



(4) RAID level-3:-

→ Use byte-level striping + single parity
→ All parity information stored in one dedicated disk
→ Data can be recovered from parity bits.
→ If parity disk crash, Data cannot be recovered.

(5) RAID level-4:-

→ Uses block level striping + parity bits.
→ All parity bits are stored in a single dedicated disk

(6) RAID level-5:-

→ Block level Striping + Single/Distributed parity.
→ Data can be recovered if any of the disk crashed, as parity bit is distributed.

(7) RAID level-6:-

→ Block level striping + Double distributed parity.
→ Use Reed Solomon code
→ More memory + costly

(B) Backup :-

→ Saving a copy of some data on another storage.
→ Bringing data from that storage back to the local disk is called Restore.

Three types :-

(i) Full Backup :-

→ Needs minimum time to store.
→ Takes max. time to create the backup.
→ More memory and bandwidth.

(ii) Differential Backup :-

→ After the last full backup, only the data that has changed is backed up.

(iii) Incremental Backup :-

→ After the last full backup, only the updated / newly added data is backed up.